HHS Enterprise Performance Lifecycle (EPLC) Framework

Role of Critical Partners:
Enterprise Architecture (EA) Critical Partner

Version 4.0
June 2018

This Enterprise Architecture Critical Partner Guidance Manual is based upon
HHS Enterprise Performance Life Cycle Framework v2.0 dated July 2012

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1. Purpose of this Manual

The purpose of this manual is to:

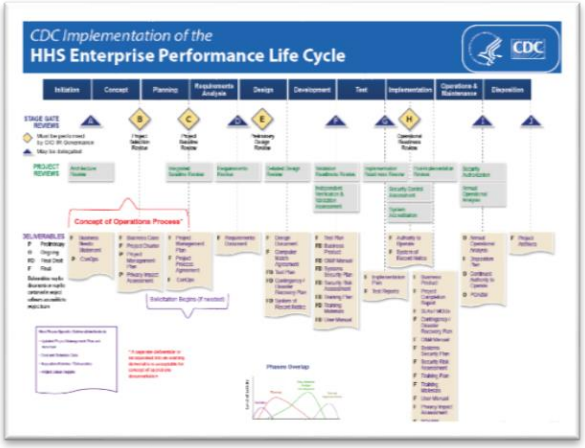
- Provide guidance to the Enterprise Architecture (EA) Critical Partner to perform the required project reviews and provide recommendations during the EPLC;
- Promote successful project completion; and
- Serve as a quick reference manual.

Beginning with section 7 of this manual, the following documentation's structure was used to partition and facilitate the critical partner's guidance for each of the phases of the EPLC:

- **Level of effort** – Estimate (low, med, high) on the amount of involvement or time needed to engage in the specified phase of the project;
- **Deliverables** – work products of each phase of the EPLC process that are available for your review to make your assessment of the project being analyzed;
- **Project Risks** – possible issues and/or risks that may arise from the lack of review at this stage;
- **Major Probing Questions** – fundamental questions, actions, and/or rationale you ought to consider during each of the EPLC's phases to perform proper project reviews.

2. About the EPLC Framework

In October 2008, HHS issued the HHS's Policy for IT Enterprise Performance Life Cycle (EPLC) which establishes a project management framework designed to adequately manage project's risk as well as project accountability. All CDC's IT projects are expected to leverage the EPLC's framework as a standard approach to perform successful IT projects. Critical Partners are an intricate part of the EPLC Framework to help perform project reviews throughout the multiple stages of a project and to share their advice and recommendations based on their respective areas of expertise/knowledge.



The picture to the right is shown for illustration purposes and represents one of multiple EPLC's diagrams and guidance available based on the type of project (small, large, etc) and effort needed. You can browse and download the available EPLC diagram(s) and related documentation online at the CDC's Enteprise IT Portfolio Office (EITPO) SharePoint site: (http://esp.cdc.gov/sites/OCOO/EITPO/EPLC/Pages/Framework.aspx?#full).

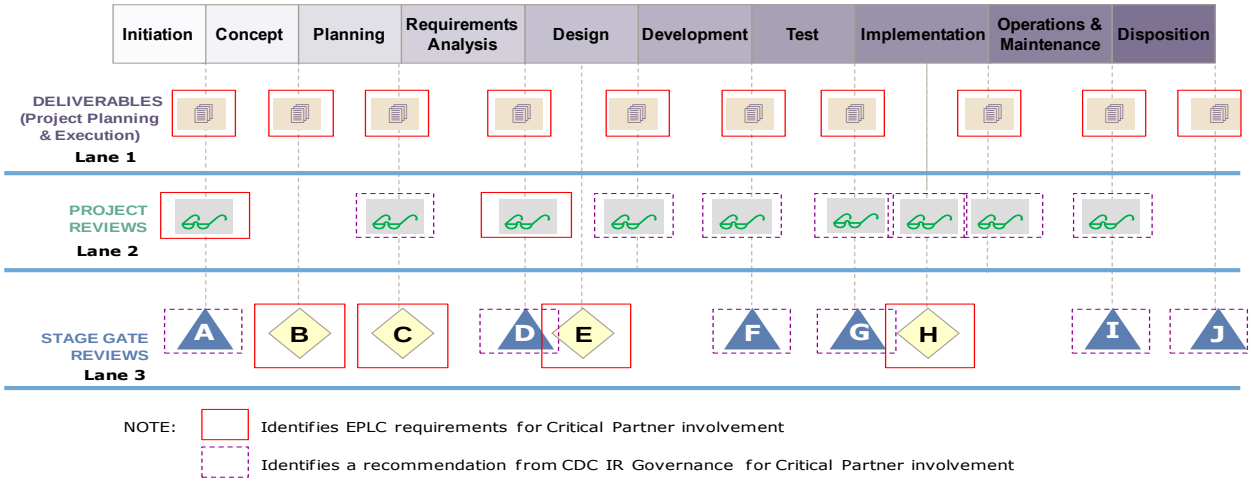
3. Overview of the Critical Partner's Role in a Project

The EPLC defines the Critical Partners as essential project stakeholders and functional managers in ten areas: Acquisition, Budget, Capital Planning and Investment Control (CPIC), Enterprise Architecture (EA), Finance, Human Resources, IT Infrastructure, IT Security, Performance, Records Management, and Section 508. In addition to these, CDC has added the IT Infrastructure Critical Partner to focus on the IT infrastructure environment and a Finance Critical Partner (combines Budget & Finance). Critical Partners are expected to participate in IT projects on an ongoing basis and provide IT governance recommendations during stage gate reviews.

The Centers, Institutes, and Offices (C/I/Os) across CDC may also define additional Critical Partner roles such as Health Scientists, Statisticians, or Epidemiologists. In this manual, the term IR Governance encompasses all organizational governance entities at CDC including those at the C/I/O level. Because of the varying organizational structures at CDC, the resource personnel and necessary expertise for these Critical Partner roles may be fulfilled by one or multiple individuals in the organization as best defined by the IR Governance.

Critical Partner Involvement in the EPLC

The EPLC framework, shown in the figure below, consists of ten life-cycle phases and three major lanes of activities that are conducted during a phase. Critical Partners have responsibilities in all phases and lanes of activities.



Lane 1: Deliverables (Project Planning & Execution)

The project manager is responsible and accountable to the Business Owner for meeting the business requirements of a project within the cost, schedule, and scope baselines. In order for a project manager to be successful, **Critical Partners need to ensure that project requirements from their respective areas are planned for at the earliest possible point in the project**. This requires that Critical Partners be **actively engaged** in the project from beginning to end. EPLC also considers Critical Partners to be members of the Integrated Project Team assisting project managers with the planning and execution of the project.

Lane 2: Project Reviews

There are 12 different project reviews that are required by the EPLC. These project reviews are conducted at specific points in the life cycle to confirm that events have occurred and decisions have been made before continuing with the project. Critical partners must leverage these project reviews to ensure they have communicated their observations with the team and can review timely responses. Some of these project reviews may be performed concurrently (e.g., the System Re-Certification and System Re-Accreditation Project Reviews in EPLC may be performed together as a part of the CDC Certification & Accreditation process).

Lane 3: Stage Gate Reviews

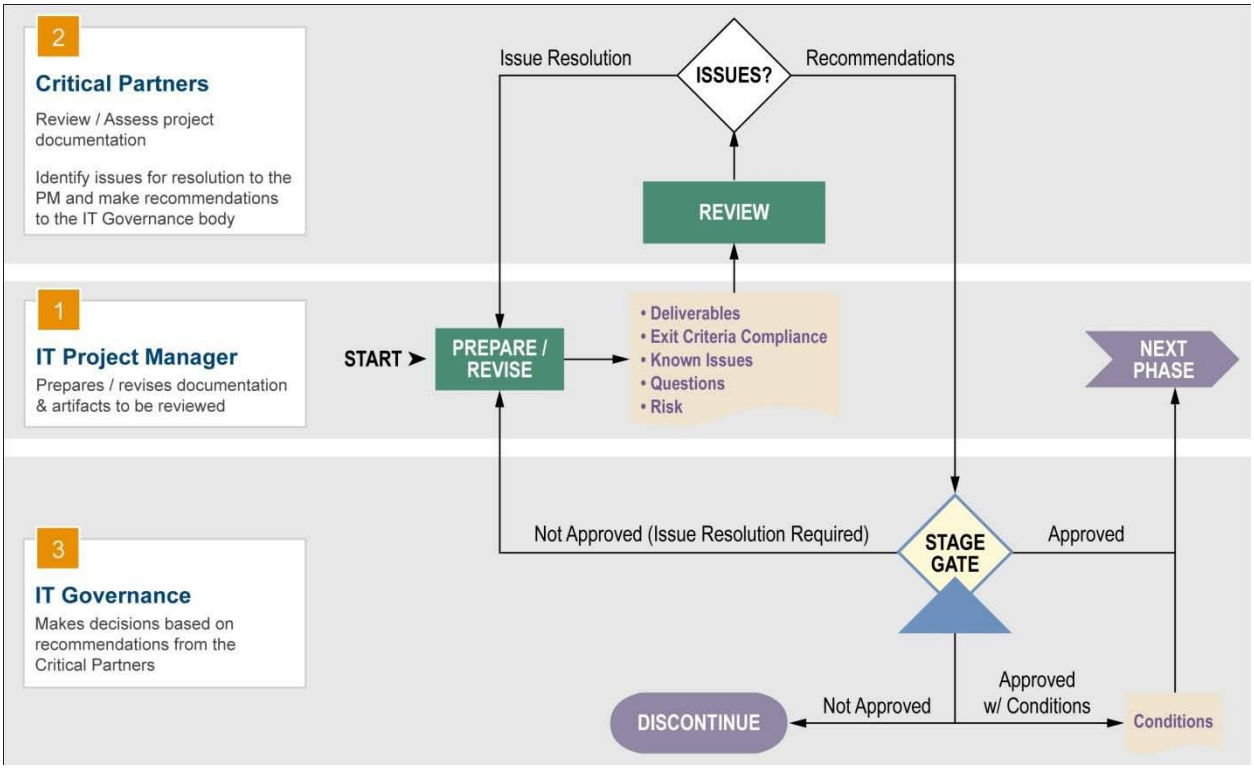
Stage Gate Reviews consist of an independent confirmation by Critical Partners to the IR Governance that Project Managers had satisfactorily produced all the required deliverables and adequately met all exit criteria for the phase to permit advancement to the next phase. Critical Partners must review

project documents for completeness, accuracy and adequacy and identify recommendations to the IR Governance regarding quality of work performed, risks for moving forward, and project's readiness to advance to the next life cycle phase. The Project Manager is also responsible for providing documentation of known issues and plans to mitigate the risks, if any.

The emphasis of Stage Gate Reviews is on:

- The successful accomplishment of the phase objectives.
- The plans for the next life cycle phase.
- The risks associated with moving into the next life cycle phase.

Stage Gate Reviews also address the availability of resources to execute the subsequent life cycle phases. The Stage Gate Review process is illustrated in the figure below.



The IR Governance may choose to approve the project's continuation to the next phase with or without conditions. Approval with conditions requires the IR Governance to establish a process for maintaining oversight of the project to ensure conditions are met. The IR Governance organization may require issue resolution by the Project Manager before approving continuation, and is responsible for discontinuing any project which fails to resolve serious issues.

Summary

Critical Partners are key project stakeholders and must be involved in all phases and activities of a project including the project planning and execution along with the appropriate reviews that occur throughout the life cycle. The EITPO's EPLC site provides detailed information regarding each phase of the EPLC and project life cycle.

The level of involvement and commitment of the Critical Partner is directly related and varies by the project, project size, purpose, risk and phase of the project. During the various phases of the project, interaction with other Critical Partners is encouraged especially where specific expertise is needed that might involve cross-functional skills.

Critical Partners:

- Ensure that IT projects are planned and implemented in a manner that meets all requirements and business needs;
- Responsible for participating in developing, reviewing and evaluating project documentation in their area of expertise;
- Ensure that there is compliance with policies in their respective areas;
- Ensure that cross-functional issues are either resolved at the staff level or articulated to the IT Governance organization for resolution;
- Participate in the making of timely tradeoff decisions when conflicts arise during the planning and execution of a project;
- Provide advice to project managers and business stewards / owners throughout the life of a project; and,
- Conduct detailed reviews for IR Governance and provide recommendations to IR Governance from Stage Gate Reviews.

4. Overall Responsibility of the Enterprise Architecture Critical Partner (EA CP)

The EA Critical Partners (EA CP) are charged with the responsibility to ensure that new solution architecture's designs consider and leverage the existing IT strategies and guidance in place. The role of the EA CP is of strategic importance as it aims to help maximize business outcomes by promoting architecture best practices which can help enable business agility, data interoperability, reusability, cost savings, and innovation. The EA Critical Partner focuses on solution architecture during the EPLC but brings wider organizational and federation-wide strategies early to the process.

Core EA CP Responsibilities:

- Minimize redundancy and silos across;
- Promote common services and cross-organizational solutions;
- Ensure completeness of solution architecture documentation (ConOps, Data Flows, Information Exchange, etc);
- Ensure that project teams discover and leverage existing organization's assets including approved business domain's architecture, domain vocabularies, common solutions, architecture blueprints, best design principles, and current IT strategies;
- Validate alignment and compliance of new proposed solution designs with federation-wide policies for Cloud first, Open Data, Open Source, reduced complexity and shared services;
- Provide technical expertise and organizational knowledge to help guide proposed product designs and/or solution architectures.

5. Reference Architectures, Design Principles, and IT Strategy

In the field of enterprise architecture, existing *Reference Architectures*, *design principles*, and *IT strategies* are fundamental tools to communicate common solution-design's guidance and best practices for architects and developers when designing new IT solutions during the EPLC. Some of the available guidance may offer common federation-wide strategies such as *Service Oriented Architecture, Microservices, Cloud, and Mobile Apps design patterns*. Other architecture guidance may exist to assist with specific CDC's business domains such as Surveillance, Laboratory, Epidemiology, Health Surveys, HR, and/or Financial and Administrative domains. Leveraging these architecture's guidances consistently and early during the EPLC helps ensure that project teams can design solutions that play well in the CDC's IT ecosystem and meet the goals and business strategies of the organization.

You can find a list of approved reference architectures, design principles, and prevailing IT strategies online by browsing your respective C/I/O's web site and/or by browsing the ***CDC Target Architecture's SharePoint site*** (see link below). All EA critical partners should become familiar with the available existing reference architectures and solution design principles promoted by the organization.

A fundamental task of the EA Critical Partner is to ensure that project teams and contractors are aware of the available reference architecture and IT solution best practices guidance and that they have considered its applicability when conceptualizing and designing new IT solutions for CDC.

6. Referenced Documents

1. **Reference Architectures, principles, and best practices** promoted by CDC's organizations and the federation - https://esp.cdc.gov/sites/ocoo/EITPO/EA/Pages/CDC-EA-Target-Architecture.aspx
2. **Common Approach to EA** – document contains guidance on a common approach for Enterprise Architects and also list the federal EA principles used through-out this manual. http://esp.cdc.gov/sites/ocoo/EITPO/EA/Reference%20Documents/common_approach_to_federal_ea.pdf
3. **Concept of Operations** – How to https://esp.cdc.gov/sites/ocoo/EITPO/EPLC/SitePages/Concept%20of%20Operations.aspx
4. **Minimal Documentation Required for all IT Systems**: https://esp.cdc.gov/sites/ocoo/EITPO/EA/Reference%20Documents/EA%20Reviews%20-%20Minimum%20Documentation%20for%20IT%20Systems.pdf
5. **CDC Business Function & Sub-function codes** can be found at: https://esp.cdc.gov/sites/ocoo/EITPO/EA/Reference%20Documents/CDC%20Business%20Functions%20and%20Sub-Functions.pdf?Web=1
6. **Architecture Review**– provides guidance on performing the required environmental scan during the EPLC's Initiation Phase to identify possible duplication or reusability opportunities: https://esp.cdc.gov/sites/ocoo/EITPO/EA/Pages/AR.aspx

7. **CDC and HHS IT Strategic Plans** can be found at: https://esp.cdc.gov/sites/ocoo/EITPO/Pages/Home.aspx

7. Initiation Phase: Role of an EA Critical Partner

The EA Critical Partner (EA CP) must understand the business need or idea being proposed and ensure that it is clearly articulated, aligns with the actual organization's strategies and goals, and is not redundant or duplicative. This requires performing an environmental scan (part of the *Architecture Review*) to determine if one or more systems and/or applications in the existing portfolio could meet the business need and should be considered as a possible alternative solution. Finally, the EA CP is charged with helping identify whether the proposed idea represents a common need across organizations and hence should be considered as a shared service.

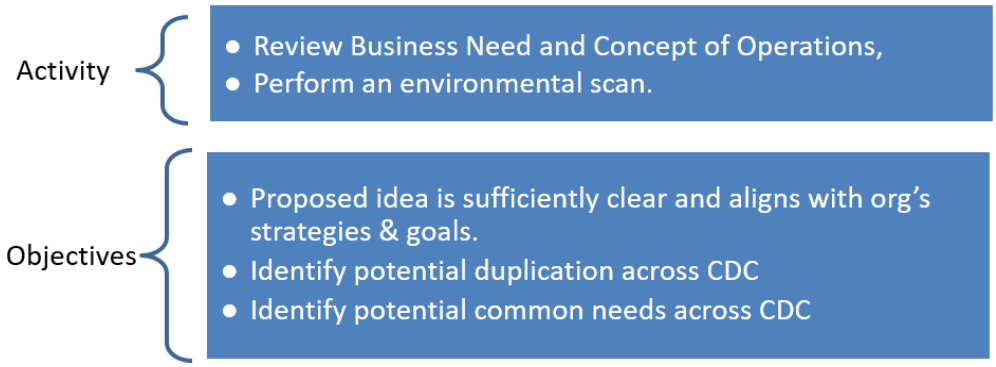


Figure 1: Initiation Phase

The following Federal Enterprise Architecture Principles (see complete list in Appendix) guides the EA Critical Partner during this phase:

- Agencies must ensure that investment decisions result in the achievement of strategic and/or tactical outcomes by employing technology and other resources in an effective manner. A number of Architecture Principles serve to ensure this occurs.
- Agency business activities exist to meet strategic objectives
- Services should be standardized within and between agencies where possible.

Enterprise Architecture (EA) Critical Partner Anticipated Level of Effort

☐Low
☐Medium
☒High

Project Risks

- The Business Needs Statement and/or ConOps might be poorly described or incomplete in terms of describing the true business needs, who the stakeholders are, and the related proposed business operations changes which can significantly raise the risk for bad resource planning and poor investment decisions.
- Redundant or overlapping solutions waste tax payer money.
- Common needs across multiple organizations or divisions should be identified early and considered for a possible shared services approach.
- An idea or business need which is poorly aligned to the organization's strategies and priorities will result in IT investments that cannot support the actual mission needs.

Review Deliverables

Note: The following documentation should be available for review by the EA CP. This list does not necessarily indicate the full extent of the expected and required documentation during this phase.

- **Business Needs Statement** (Final): The Business Needs Statement includes a brief description of the proposed project's purpose, goals, and scope. Sufficient high-level functional requirements are necessary to understand what the project is intended to do and how it supports the business need.
- ***Concept of Operations (preliminary)***, also known as **ConOps**, identifies the intended end-users and describes their respective proposed business operations which should help define the overall scope from a business operations perspective.

During the Initiation phase, an enterprise architecture review, also known as "Architecture Review" is expected. This review aims to ensure that the business need or idea is clearly stated, doesn't duplicate what already exist, and considers reusing existing IT assets. The following major probing questions for the EA CP essentially help you perform an Architecture Review, but additional and more detailed documentation can be found online. Please see *Referenced Document*.

Major Probing Questions for the Enterprise Architecture (EA) Critical Partner

Use the following to assist you with completing your responsibilities for this phase as appropriate to your subject matter expertise.

Question	**Objective Evaluation Considerations**
1. *Is the Business Needs Statement sufficiently clear and identifies the target business functions and operations in scope?*	Read the Business Need Statement and identify the main business functions and/or business operations that will be impacted with the proposed change or idea, at least at a high level. Otherwise, the BNS will be too vague to proceed. The Business Needs Statement should typically not identify specific technical solutions or vendors unless the stated need or driver describes a need to refresh/update the technology for an existing IT solution. Review any preliminary Concept of Operations (ConOps) type documentation to aid with clarifying in-scope business operations and impacted stakeholders.
2. *The degree to which the need and idea for change aligns to your organization strategic plan and the CDC IT Strategic plan?*	Validate that there exists a clear line-of-sight to one or more elements of a relevant strategic plan. A lack of such alignment would suggest this project is not a good candidate for moving forward.
3. *Is there another project with similar business needs in the works? Or, is there another IT solution in production that could potentially meet the identified business needs?*	Review the current IT portfolio catalog(s) to determine the existence of systems that can possibly support this business need. In addition, determine if other projects are underway or being considered to address this or a similar need. Consult the online *Architecture Review* documentation for additional information on how to search the IT portfolio catalog(s).

4. *Could the identified business needs be common across other organizations? I.e., is there an opportunity to consider shared services and the larger need?*	Consider whether the proposed idea or business need may be a common need across multiple organizations. Propose the possibility and opportunity to get additional collaboration and funding to address a broader organizational need as a shared service.

8. Concept Phase: Role of an EA Critical Partner

The Concept Phase should provide you and all stakeholders with a much more complete view of the proposed business change idea and possible solution's paths (i.e., alternative solutions). During this stage, multiple business processes and/or scenarios should be considered rather earlier to explore ways for improving or simplifying the current business processes and gain efficiencies. In addition, multiple technical solution's alternatives ought to be explored that can reduce the overall complexity, cost of ownership, and achieve business efficiencies. A goal for the EA CP in this phase is to influence the decision making process that leads to the discovery of alternative solutions, making sure that any existing organizational IT strategies are considered early by the team. Such IT strategies, which follows HHS and federation-wide guidance in place, may include the need to first identify possibilities for Cloud services and/or shared services, enabling data interoperability (APIs, etc), and leveraging any approved reference architecture(s). These strategies will influence the discovery efforts for solution alternatives.

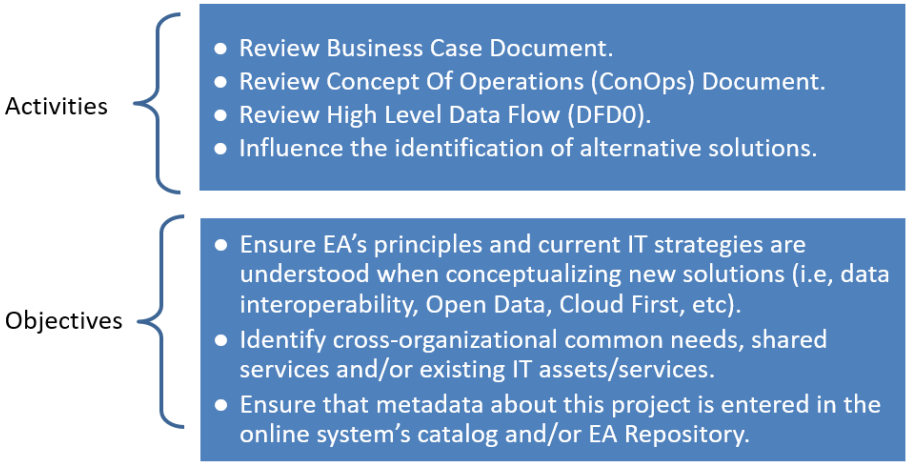


Figure 2: Concept Phase

Federal Enterprise Architecture principles that especially apply to this phase:

- Agencies must ensure that investment decisions are based on architectural solutions that result in the achievement of strategic and/or tactical outcomes by employing technology and other resources in an effective manner
- Agency business activities exist to meet strategic objectives
- Services should be standardized within and between agencies where possible
- Services should be web-enabled whenever possible
- Data and information exchange should be based on open standards
- Open-source software solutions should be included in alternatives analyses
- Use cloud-based application, platform, and infrastructure hosting designs whenever possible to promote scalability, cost-efficiency, and metering
- Public information must be made accessible, discoverable, and usable by the public in a secured way

Enterprise Architecture (EA) Critical Partner Anticipated Level of Effort

☐Low
☐Medium
☒High

Review Deliverables

Note: The following documentation should be reviewed by the EA CP. This list does not necessarily indicate the full extent of the expected and required documentation during this phase.

- **Project Charter** (Final): The Project Charter formally authorizes a project, describes the business need for the project, and the product to be created by the project. It provides the project manager with the authority to apply up to a certain level of organizational resources to project activities.
- **Business Case** (Final): A Business Case is a documented, structured proposal for business improvement that is prepared to facilitate a selection decision for a proposed project by organizational decision makers. It identifies the high-level requirements that are to be satisfied, an analysis of proposed alternative solutions (with reasons for rejecting or carrying forward each option), assumptions, constraints, a risk-adjusted cost-benefit analysis, and preliminary acquisition plan.
- **Concept of Operation (ConOps) Document** – A *Concept of Operations (ConOps)* describes the future state business operations in the terminology of its end-users. The ConOps is written in non-technical and easy-to-understand manner to capture consensus and/or agreement from the business users on the scope and how their business operations would change with the proposed solution or idea**.** A basic template document for ConOps can be downloaded at: https://esp.cdc.gov/sites/ocoo/EITPO/EA/Reference%20Documents/CONCEPT%20OF%20OPERATIONS.pdf
- **High Level Data Flow Diagram (DFD) or Information Data Exchange** (Preliminary) – A conceptual high level data flow diagram ought to be produced to indicate what data is expected to flow, and where, in relation to the identified business operations. Essentially, it is high level at this point, but should identify data entities and where data needs to flow in and out of the system and to/from other external systems. This document ought to be reviewed at this stage as it can help you identify architectural requirements and IT strategies that need to be considered as we look for alternative solutions (i.e., data interoperability needs, possible APIs, additional security controls, etc). Note: This information regarding data flows and data exchanges may also be included as integral part of the *ConOps* document.

The EA CP must infer strategic and tactical architectural requirements from the documentation produced and must communicate with the team to help influence the process of finding alternative solutions but also possibly influence project planning. The project team must make accessible the documentation online, leveraging existing CDC collaboration tools such as SharePoint.

Project Risks

- Missing or incomplete ConOps could lead to poor consensus on scope and initial requirements which will negatively impact success in later phases.
- An inadequate preliminary alternatives analysis could lead to missing entire categories of best potential solutions.
- The need for transmitting or receiving data and information across other organizations and systems may be obviated or identified too late and after the project plan and budget are estimated. This will

results in such estimates missing considerations for key architectural requirements such as possible service interfaces and security controls required.

- Failure to identify appropriate reference architectures, vocabulary and/or organizational data standards can lead to a solution design that does not comply with the organization's IT strategies, or with the existing infrastructure ecosystem, or cannot interoperate with other systems in place.
- Failure to identify existing systems or planned projects that satisfy some or all of the business needs stated for the project can result in duplicate IT capabilities increasing overall IT cost.

Major Probing Questions for the Enterprise Architecture (EA) Critical Partner

Use the following questions to assist you with completing your responsibilities for this phase as appropriate to your subject matter expertise.

Question	**Objective Evaluation Considerations**
1. *Is the project scope, as expressed in the Business Case and/or Concept of Operations, clear and well aligned with the strategy and goals of the organization?*	Validate that scope has been expressed in such a way that stakeholders would have been able to understand and come to agreement regarding it.
2. *Is there a Concept of Operations (ConOps) or similar documentation that has captured the consensus of the affected business stakeholders regarding future state operations and the nature of desired changes?*	The ConOps or a similar description of proposed business operational changes is needed to begin evaluating scope and possible data integration needs. Leverage the ConOps to begin identifying any significant architecture requirements (points for data exchanges, common services, cross-organizational efforts, etc) which would influence the discovery of alternative solutions during this stage.
3. *Has the need for sending and receiving information with other processes and systems been identified - at a high level - to begin discerning corresponding necessary interfaces and security controls?*	Review the available documentation to aid in the identification of potential data exchange points or interfaces across organizations and/or external parties. Continue identifying significant high-level architectural requirements like points for data interfaces (API), common transport protocols, security considerations, web services, etc. This information is crucial to inform the discovery of alternatives solutions and subsequent EPLC phases.
4. *Were alternative approaches and solutions properly considered?*	Review documentation related to the analysis of alternatives conducted. Participate in that analysis as required. Were all alternatives properly considered – e.g., status quo, leverage/enhance existing solutions, government owned and operated solution, commercial, build.
5. *Has the team considered existing reference architectures, shared services, and/or relevant vocabularies in place?*	Ensure the project team has considered any existing architecture plans, IT standards and strategies such as leveraging Cloud and shared services. Also, help identify any common IT assets (software or services) in the organization that could be considered for reuse by the solution (i.e., COTS, GOTS, shared services, etc). Your input will help the team ensure that decisions during the discovery and selection of alternative solutions can minimize redundancies

	and align with organizational IT strategies and goals.

9. Planning Phase: Role of an EA Critical Partner

As a Critical Partner, get involved early in this phase to aid the project team with addressing and planning any cross-organization considerations. For instance, collaboration with other internal/external organizations and teams must be planned and strategized as IT solutions become shareable across organizations and to prevent further silos. Additional stakeholders in other organizations may also need to be interviewed and accounted for in the project plan. The Collaborative Planning Methodology (CPM) of the *Common Approach for Federal EA* (see Referenced Documents) describes step-by-step instructions and best practices on collaboration and how/when the stakeholders need to be brought in.

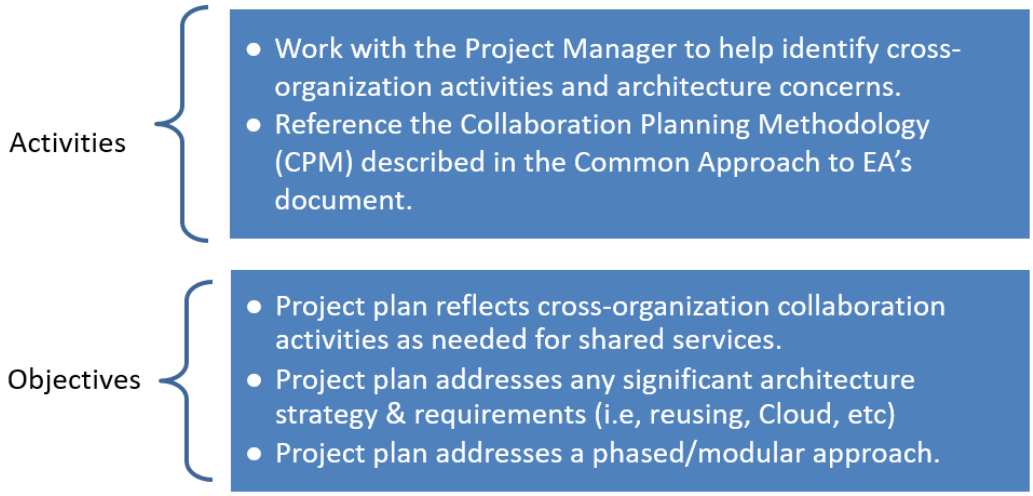


Figure 3: Planning Phase

Work with the project manager to help identify and agree on such collaboration steps as input to the project plan and for leveraging any reusable resources such as business components and shared services across organizations. Make sure that the project plan has included the production of the necessary artifacts and deliverables needed for subsequent phases and stage gate reviews.

Federal Enterprise Architecture principles that must be observed during this phase:

- Agencies must ensure that investment decisions are based on architectural solutions that result in the achievement of strategic and/or tactical outcomes by employing technology and other resources in an effective manner
- Agency business activities exist to meet strategic objectives
- Services should be standardized within and between agencies where possible
- Services should be web-enabled whenever possible
- Data and information exchange should be based on open standards
- Open-source software solutions should be included in alternatives analyses
- Use cloud-based application, platform, and infrastructure hosting designs whenever possible to promote scalability, cost-efficiency, and metering

Enterprise Architecture (EA) Critical Partner Anticipated Level of Effort

☐Low

☒Medium

☐High

Review Deliverables

Note: The following documentation should be available for review by the EA CP. This list does not necessarily indicate the full extent of the expected and required documentation during this phase.

- **Project Management Plan** (Final): The Project Management Plan is a formal, approved document that defines the overall plan for how the project will be executed, monitored and controlled.
- **Project Process Agreement** (Final): A Project Process Agreement (PPA) is used to authorize and document the justifications for using, not using, or combining specific EPLC stage gate reviews and the selection of specific deliverables applicable to the project, including the expected level of detail to be provided. Completion of a PPA takes place in the planning phases of the project life cycle and may be included as part of the Project Management Plan (PMP). Whether part of the PMP or a standalone document, the completed PPA should be signed by the project manager/sponsor. Once approved by an IT governance organization this document authorizes the project to proceed according to agreed-upon EPLC approach documented within the PPA document.

Project Risks

- The project plan may be missing necessary planning activities for cross-organizational efforts and/or other related architectural requirements identified in the previous phases.
- Plans need to include activities/time associated with creating critical artifacts and collaboration with cross-organization teams.

Major Probing Questions for the Enterprise Architecture (EA) Critical Partner

Use the following questions to assist you with completing your responsibilities for this phase as appropriate to your subject matter expertise.

Question	**Evaluation Considerations**
1. *Has anything of significance changed from previous phase review?*	Projects move forward through progressive elaboration. Review any pending responses for observations made in previous phases and ensure that agreed upon changes to scope were properly reflected in the ConOps and Business Case documentation.
2. *Have all key EA CP's concerns and observations previously submitted been addressed?*	Request any updates to the business case, Concept of Operations, and/or other documentation that addresses change in scope. Request access to planning related deliverables. Confirm that each significant architecture concern/observation (e.g., points of integration, cross-organizations concerns, redundancy, reference architectures, etc.) have been appropriately addressed; most importantly in the actual schedule and budget.
3. *Does the project plan account for identified*	Confirm that the schedule has included tasks for

cross-organizational activities and dependencies. E.g., Planned use for shared services, interfaces, etc.	engaging with other organizations and parties involved in the scope of this project.
4. *Are there plans for a phased and/or modular approach?*	Consider the full "product" scope and consider possibilities for further partitioning, a more modular approach.

10. Requirements Phase: Role of an EA Critical Partner

During the requirements phase and requirements review, determine if the requirements also account for all of the architectural requirements you had identified and discussed in previous phases. Identify and relay to the team where elements of the agreed architectural IT strategies discussed may have been ignored or were poorly assessed for requirements. For instance, architectural requirements that had identified existing vocabularies and/or referenced architectures to use, or the need for leveraging mobile devices and Cloud, all may carry additional requirements gathering steps that must have been identified in this phase. In addition, any identified functional commodity (such as existing shared services) would require that additional stakeholders and/or cross-organizational entities get included in the requirements collection phase. Continue efforts to identify areas of common cross-organizational needs that may have surfaced during the requirements gathering process and that may possibly translate to new architectural requirements such as the need for leveraging shared services. Verify that logical data and information exchanges are modeled as part of the requirements document. Finally, establish that the Business Process Models and Logical Data Models are documented at the proper level.

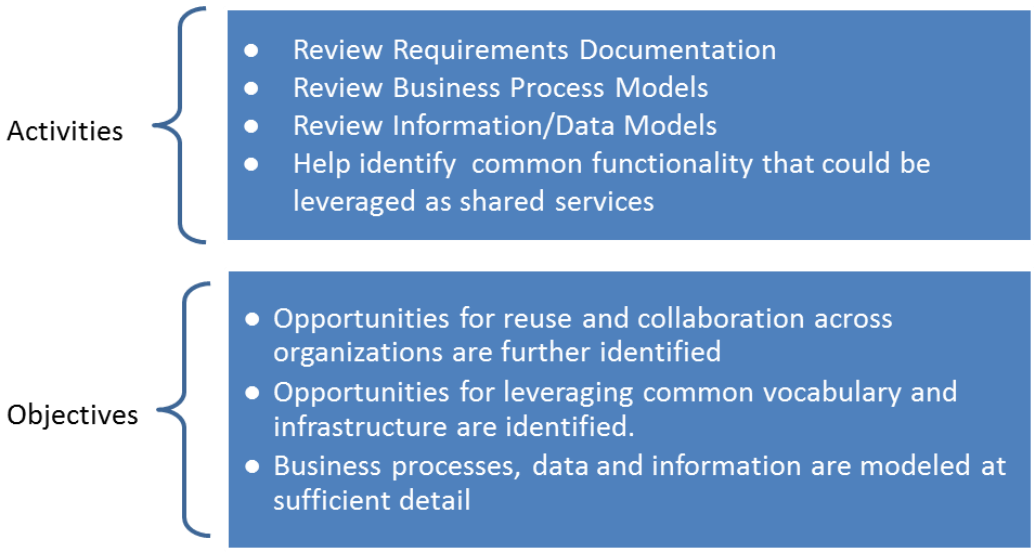


Figure 4: Requirements Phase

Federal Enterprise Architecture principles that especially apply to this phase:

- Agencies must ensure that investment decisions are based on architectural solutions that result in the achievement of strategic and/or tactical outcomes by employing technology and other resources in an effective manner
- Agencies should select reusable and sharable services and products to obtain mission or support functionality.
- Agency business activities exist to meet strategic objectives
- Services should be standardized within and between agencies where possible

- Services should be web-enabled whenever possible
- Security controls must be designed into IT support of every business process
- Data and information exchange should be based on open standards
- Privacy considerations must be designed into every data solution
- Use well documented interfaces built on non-proprietary open platforms using standard platform independent data protocols (such as XML)
- Open-source software solutions should be included in alternatives analyses
- Use cloud-based application, platform, and infrastructure hosting designs whenever possible to promote scalability, cost-efficiency, and metering
- Security controls must be designed into every technology solution
- Public information must be made accessible, discoverable, and usable by the public in a secured way

Enterprise Architecture (EA) Critical Partner Anticipated Level of Effort

☐Low
☐Medium
☒High

Review Deliverables

Note: The following documentation should be available for review by the EA CP. This list does not necessarily indicate the full extent of the expected and required documentation during this phase.

- **Requirements Document** (Final): The Requirements Document provides validation and further analysis of the business requirements that were documented in the earlier phases. It defines the functional requirements, non-functional requirements, the business model, and logical data model. The complete set of requirements serves as a blueprint for the acquisition planning and subsequent design activities.

In addition to the Requirements Document, consider the following artifacts that should be available for your review:

- **Business Process Models** documented at the proper level.
- **Logical Data Models** documented at the proper level and include logical data entities as well as data flow diagrams that can identify data transfers and exchanges to and from outside partners.

Project Risks

- An incomplete or insufficiently detailed business process model can lead to an incomplete set of requirements. Incomplete requirements will lead to performance gaps in the final system.
- Failure to document relevant architectural details diminish the ability to ensure the solution will align with the Enterprise Architecture.

Major Probing Questions for the Enterprise Architecture (EA) Critical Partner

Use the following questions to assist you with completing your responsibilities for this phase as appropriate to your subject matter expertise:

Question	**Evaluation Considerations**
1. *Are all key elements of scope and stakeholders, as identified and confirmed in previous phases,*	Request the most current version of the Business Case, Concept of Operations, and Requirements documentation and look for major operational

adequately represented in the requirements?	gaps and/or inconsistencies across these documents. Key business operations described in the Requirements Document should trace to operations identified in the ConOps and/or Business Case documents. Any gaps found should be discussed as it could represent incomplete requirements and/or unexpected changes to the business scope for the project.
2. *Have all the architectural requirements/observations identified in previous phases been addressed/included in the actual requirements document?*	Architectural requirements identified in previous phases should be included/addressed in the latest Requirements Document. Reference architectures, domain vocabularies, cross-organizational needs (i.e., shared services), and/or data/information standards are examples of architectural requirements. Review your architectural requirements findings from previous phases and reconcile anything that seems to be unaddressed in the latest requirements document.
3. *How complete are the Business Process models, Information Exchange Data Flows, and Logical Data Models?*	Request and review these key models – they should exist and be well-defined at this stage.

11. Design Phase: Role of an EA Critical Partner

The outcome of the Design Phase is the completion of a detailed technical design to guide the development efforts. During the design phase and before the design project's review has been completed, confirm that the minimum recommended design artifacts (see Deliverables) were produced and are made accessible online from a central repository or SharePoint site. Conduct a formal review of the high-level architectural design so that you are confident that the design satisfies the business requirements and leveraged prescribed EA principles for interoperability, shared services, and Cloud as applicable.

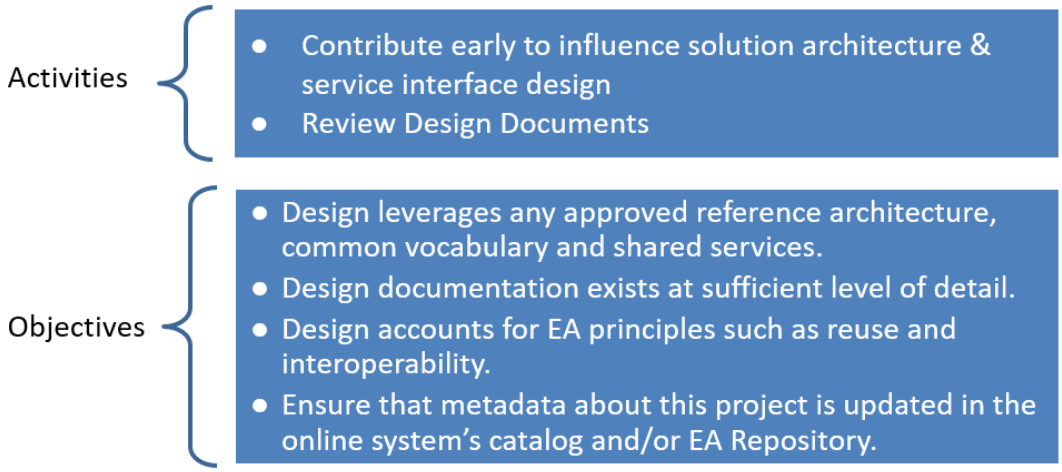


Figure 5: Design Phase

Federal Enterprise Architecture principles that especially apply to this phase:

- Agencies must ensure that investment decisions are based on architectural solutions that result in the achievement of strategic and/or tactical outcomes by employing technology and other resources in an effective manner
- Agencies should select reusable and sharable services and products to obtain mission or support functionality.
- Services should be standardized within and between agencies where possible
- Services should be web-enabled whenever possible
- Security controls must be designed into IT support of every business process
- Data and information exchange should be based on open standards
- Privacy considerations must be designed into every data solution
- Use well documented interfaces built on non-proprietary open platforms using standard platform independent data protocols (such as XML)
- Open-source software solutions should be included in alternatives analyses
- Use cloud-based application, platform, and infrastructure hosting designs whenever possible to promote scalability, cost-efficiency, and metering
- Convergence in voice, data, video, and mobile technologies supports infrastructure consolidation, which should be pursued wherever possible
- Host solutions must be compliant with federal policy and standards (e.g., Trusted Internet Connection, IPv6 routing, and PIV authentication)
- Desktop/mobile solutions must be compliant with the latest US Government Configuration Baseline standard
- Security controls must be designed into every technology solution
- Public information must be made accessible, discoverable, and usable by the public in a secured way

Enterprise Architecture (EA) Critical Partner Anticipated Level of Effort

☐Low
☐Medium
☒High

Review Deliverables

Note: The following documentation should be available for review by the EA CP. This list does not necessarily indicate the full extent of the expected and required documentation during this phase.

❖ **Design Documents**: Based on EA best practices, the following set of documentation must exist to support a robust solution design for the organization. The typical development team may include additional documentation but the ones listed here are the minimum required at the stage gate review. Validate that the following documentation exists with a sufficient level of detail. A link with more information and/or examples for these required standard documents can be found in the *Referenced Document* section of this document:

Architecture Domain	Core Artifact
Business	High-Level Process Diagrams
Data	Logical Data Model / Data Flows
Application	Application Component-Based Architecture
Infrastructure	High Level Network Diagram
Security	Control List

! Each of these architecture design documents are critical to understand and communicate the significant design aspects of the solution. The project team must ensure that these documents are

produced and made accessible online using the existing collaboration tools such as SharePoint.

Project Risks

Overly complex systems (solutions) designs and/or architectures can lead to solutions that are hard to maintain, fix, and/or change quickly but also raises the probability of project failure. Designs that miss major business operations and requirements will not meet the business needs and can be very costly to fix down the road.

Major Probing Questions for the Enterprise Architecture (EA) Critical Partner

Use the following questions to assist you with completing your responsibilities for this phase as appropriate to your subject matter expertise:

Question	**Evaluation Considerations**
1. *Were the recommended minimum set of "core" artifacts/deliverables fully created and can they convey the solution design clearly?*	Request access to design related deliverables. Confirm the following are documented and with sufficient detail: • Business - High-level business process flows • Data - Conceptual/logical data models and flows • Application – Detailed Application structure (component-based) and interfaces • Infrastructure – High level network diagram • Security - Controls
2. *Does the design provide adequate traceability to the business need, desired concept of operations, and requirements? I.e., are all key elements of project scope, as identified and confirmed in previous phases, adequately represented in the design?*	Confirm that the current identified business needs, major requirements (functional and non), and major key stakeholder/user's need were appropriately accounted from a design perspective. Validate that the proposed architecture/design meets the agreed future state of business operations. Confirm that alternatives designs (Open source, cloud, etc.) were adequately considered prior to selecting this solution architecture.
3. *Is the design leveraging existing architecture blueprints, principles, and standards?* *i.e., does the design address Open Data, interoperability, modularity, loosely-coupled components, and other architecture's best practices ?*	Ensure the design addresses: • Interoperability by leveraging Open Data and open source standards (e.g. HL7, ICD-10, etc.) represents a significant federation-wide's strategy. • Any existing architecture best design adopted by the organization, such as domain vocabularies, service oriented architecture design and/or Microservices, etc. • Implementing security & privacy controls in accordance with CDC Security Standards • Usability, performance and "-ilities"

	(reliability, agility, maintainability, etc.). • Minimizes implementation complexity and total costs of ownership.

12.Development Phase: Role of an EA Critical Partner

During Development Phase of the project, the EA CP serves mostly as a technical advisor to the team and based on the actual EA CP's technical skills and experience. The EA CP can provide advice on matters related to understanding aspects of the approved solution's design as well as helping identify what common organizational capabilities and/or technical standards are in-use by the organization for developing IT solutions such as standard authentication methods, mobile stores, API gateways, and any standard source-code repository. In addition, the EA CP should ensure developers ultimately store the source code produced in the available online repositories. Verify that any approved change request in this phase remain compliant with the solution's design and was documented properly.

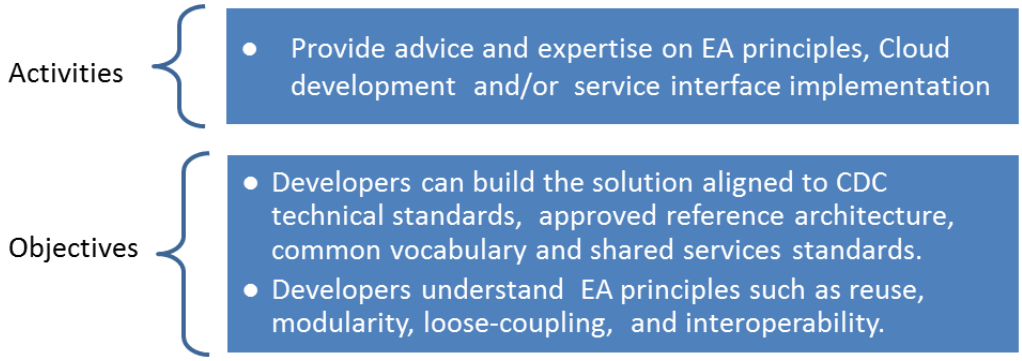


Figure 6: Development Phase

Federal Enterprise Architecture principles that especially apply to this phase:

- Agencies should select reusable and sharable services and products to obtain mission or support functionality.
- Services should be standardized within and between agencies where possible
- Services should be web-enabled whenever possible
- Data and information exchange should be based on open standards
- Privacy considerations must be designed into every data solution
- Use well documented interfaces built on non-proprietary open platforms using standard platform independent data protocols (such as XML)
- Application platforms should be virtualized whenever possible
- Use cloud-based application, platform, and infrastructure hosting designs whenever possible to promote scalability, cost-efficiency, and metering
- Convergence in voice, data, video, and mobile technologies supports infrastructure consolidation, which should be pursued wherever possible
- Host solutions must be compliant with federal policy and standards (e.g., Trusted Internet Connection, IPv6 routing, and PIV authentication)
- Desktop/mobile solutions must be compliant with the latest US Government Configuration Baseline standard
- Security controls must be designed into every technology solution

Enterprise Architecture (EA) Critical Partner Anticipated Level of Effort

☒Low

☐Medium
☐High

Review Deliverables

Note: The following documentation should be available for review by the EA CP. This list does not necessarily indicate the full extent of the expected and required documentation during this phase.

- **Test Plan** (Final): The Test Plan describes the test cases and test environment specifications, and includes a Requirements Traceability Matrix that maps requirements to the specific tests to be conducted in the Test Phase. This final draft Test Plan will be used in the Development Phase to test components as they are built and integrated. This ensures that all test cases will be adequately evaluated, executed, and system tested to ensure requirements are met.
- **Technical Documentation:** The following minimum documentation should be maintained as changes occur during the Development phase. Doing so will help to ensure the interoperability and share-ability of services as well as the efficient and effective operation and maintenance of the solution.

Architecture Domain	Core Artifact
Source-code	Well commented set of source code.
Data	Physical Data Models / Data Flows
Application	Application Component-Based Architecture
Infrastructure	High Level Network Diagram
Security	Control List implementation

Project Risks

Changes can lead to "scope creep" and a gradual move away from the agreed upon architecture design including the introduction of overlap or redundancy with the functions of other systems.

Major Probing Questions for the Enterprise Architecture (EA) Critical Partner

Use the following questions to assist you with completing your responsibilities for this phase as appropriate to your subject matter expertise.

Question	**Evaluation Considerations**
1. *Did development follow the agreed upon design? Does that design provide adequate traceability to requirements and address the business need and future concept of operations?*	Ensure what is being developed conforms to the agreed upon design in previous phase and provides traceability to the requirements and the business need
2. *Has the minimum required technical documentation been created, reflective of changes introduced during the development phase?*	Ensure appropriate technical documentation is being made available to support future maintenance activities and/or enhancements.
3. *Are all key elements of scope, requirements, and design adequately represented in the plans for testing?*	Confirm that each major project element has been appropriately considered from a development and testing perspective.

13.Test Phase: Role of an EA Critical Partner

At the discretion of the EA CP, there is very little or no involvement in the test phase. It all depends on the size of the project, the number of cross-organization teams working together, and the level of criticality of the system. The Test phase includes the thorough testing and auditing of the Business Product's design, coding and documentation.

Testing has three functions:

1. It allows you to determine how new applications will perform in a production environment
2. It verifies how well redundancy will work in a failure state
3. It helps determine how well your team members react to problems

Assist the application architect where needed in creating a checklist of core systems and backup systems that require testing. Core systems will be apparent from the documentation the team kept. Core systems will be the hardware and software without which the organization cannot exist. Often, such systems consist of the critical servers, the network backbone, and messaging pieces such as e-mail and other areas of unified communication.

Enterprise Architecture (EA) Critical Partner Anticipated Level of Effort

☒Low
☐Medium
☐High

Review Deliverables

Note: The following documentation should be available for review by the EA CP. This list does not necessarily indicate the full extent of the expected and required documentation during this phase.

- **Test Reports** (Final): Test reports should be created at the end of the testing phases to document the overall test results. These reports summarize the testing activities that were performed and describe any variances between the expected test results and the actual test results and includes identification of unexpected problems and/or defects that were encountered.
- **Implementation Plan** (Final): The implementation plan outlines activities necessary to ensure that the project's product is available for use by its end-users as originally planned. The implementation plan should present a clear alignment between regulations, policies, business and processes, and implementation objectives while clearly outlining any project constraints and/or assumptions. It will also include items such as the selection of hardware, software, and vendors; building testing technology requirements and networks, installing and testing the actual system; performing final system and user acceptance testing; preparing and implementing data conversion/migration plans; training the users; and ultimately converting existing processes and operations over to the new system.

Project Risks

Improper or inadequate testing will negatively affect the quality and efficiency of the system.

Major Probing Questions for the Enterprise Architecture (EA) Critical Partner

Use the following questions to assist you with completing your responsibilities for this phase as appropriate to your subject matter expertise.

Question	**Evaluation Considerations**
1. *Are all key elements of project scope, as identified and confirmed in previous phases, adequately represented and addressed in the plans for testing? Does the test plan provide adequate traceability to design requirements and how it addresses the business need?*	Confirm that each in-scope item, each major requirement, and each stakeholder need has been appropriately considered from a testing perspective.

14.Implementation Phase: Role of an EA Critical Partner

The role of the EA CP can be minimal during the implementation phase; although, the EA CP may serve as an advisor in matters related to deployment objectives based on the EA CP's line of expertise. The process of implementation is synonymous with installation, deployment, rollout, or go-live. Any data conversion requirements, phased-approach implementation, and/or training for using, operating, and maintaining the system must be accomplished during this phase. At a minimum, the EA CP must ensure that the system has been well documented and the documentation is made accessible.

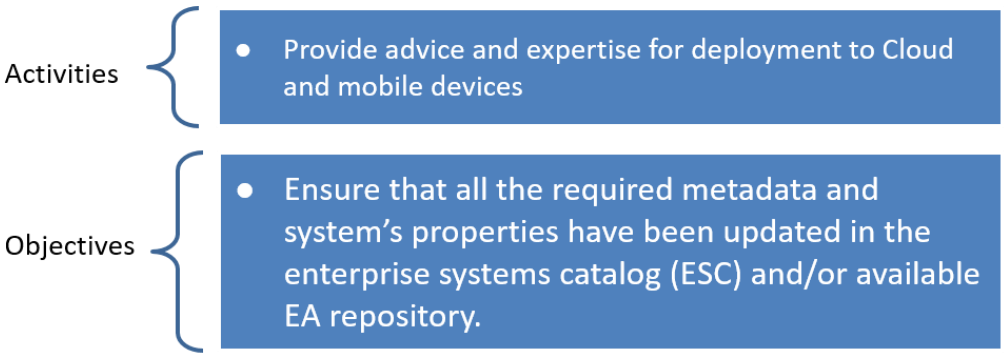


Figure 7: Implementation Phase

Federal Enterprise Architecture principles that especially apply to this phase:

- Agencies should select reusable and sharable services and products to obtain mission or support functionality.
- Services should be standardized within and between agencies where possible
- Services should be web-enabled whenever possible
- Data and information exchange should be based on open standards
- Privacy considerations must be designed into every data solution
- Use well documented interfaces built on non-proprietary open platforms using standard platform independent data protocols (such as XML)
- Application platforms should be virtualized whenever possible
- Use cloud-based application, platform, and infrastructure hosting designs whenever possible to promote scalability, cost-efficiency, and metering
- Convergence in voice, data, video, and mobile technologies supports infrastructure consolidation, which should be pursued wherever possible
- Host solutions must be compliant with federal policy and standards (e.g., Trusted Internet Connection, IPv6 routing, and PIV authentication)

- Desktop/mobile solutions must be compliant with the latest US Government Configuration Baseline standard
- Security controls must be designed into every technology solution

Enterprise Architecture (EA) Critical Partner Anticipated Level of Effort

☒Low
☐Medium
☐High

Review Deliverables

Note: The following documentation should be available for review by the EA CP. This list does not necessarily indicate the full extent of the expected and required documentation during this phase.

- **Project Completion Report** (Final): The project completion report finalizes all project activities completed across all phases of the project to formally complete the project and transfer the completed or cancelled project as appropriate. The purpose of project completion is to assess the project, ensure completion, and derive any lessons learned and best practices to be applied to future projects.

- **Technical Documentation:** The technical documentation should be finalized and placed in the appropriate project library to help ensure the ongoing interoperability and share-ability of services as well as the efficient and effective operation and maintenance of the solution going forward.

Project Risks

Changes can lead to "scope creep" and a gradual move away from the agreed upon architecture including the introduction of overlap or redundancy with the functions of other systems.

Major Probing Questions for the Enterprise Architecture (EA) Critical Partner

Use the following questions to assist you with completing your responsibilities for this phase as appropriate to your subject matter expertise:

Question	**Evaluation Considerations**
1. *Does the technical documentation accurately reflect what was implemented? Is the O&M documentation adequate for the purpose?*	Review the core technical documentation (i.e. Logical data models, process diagrams, network diagrams, etc.) as well as O&M documentation for completeness, accuracy, and usefulness.

15.Operations & Maintenance Phase: Role of an EA Critical Partner

During the Operations & Maintenance (O&M) Phase, the certified and accredited Business Product operates in full-scale production environment for sustained use and operations/maintenance support. The O&M Phase operates in a continuum and requires that an operational analysis (OA) be performed

every year to validate that the Business Product solution continues to meet the business needs and that the current technology remains updated and well-supported.

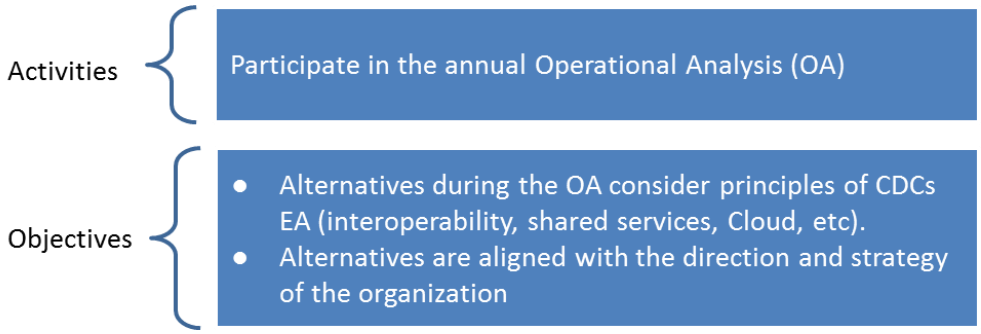


Figure 8: O&M Phase

The role of the EA CP during this phase is to contribute to the evaluation of system performance, and the degree to which it continues to be aligned with the direction and strategy of the organization and with key Enterprise Architecture guidelines. Also, the role of the EA CP is to assure the analysis of alternatives performed during the OA recommends solutions that are compliant with EA principles and aligned to any available reference architecture. The EA CP should determine if there exists an acceptable level of current state documentation.

Federal Enterprise Architecture principles that especially apply to this phase:

- Agencies must ensure that investment decisions are based on architectural solutions that result in the achievement of strategic and/or tactical outcomes by employing technology and other resources in an effective manner
- Agencies should select reusable and sharable services and products to obtain mission or support functionality.
- Services should be standardized within and between agencies where possible
- Services should be web-enabled whenever possible
- Data and information exchange should be based on open standards
- Privacy considerations must be designed into every data solution
- Use well documented interfaces built on non-proprietary open platforms using standard platform independent data protocols (such as XML)
- Application platforms should be virtualized whenever possible
- Use cloud-based application, platform, and infrastructure hosting designs whenever possible to promote scalability, cost-efficiency, and metering
- Convergence in voice, data, video, and mobile technologies supports infrastructure consolidation, which should be pursued wherever possible
- Host solutions must be compliant with federal policy and standards (e.g., Trusted Internet Connection, IPv6 routing, and PIV authentication)
- Desktop/mobile solutions must be compliant with the latest US Government Configuration Baseline standard
- Security controls must be designed into every technology solution
- Public information must be made accessible, discoverable, and usable by the public in a secured way

Enterprise Architecture (EA) Critical Partner Anticipated Level of Effort

☐Low

☐Medium

☒High

Review Deliverables

- **Annual Operational Analysis** (Final): OMB and HHS require that all steady-state projects, including the steady-state portion of Mixed Life Cycle investments, be reviewed at least annually to document their continued effectiveness in supporting mission requirements and minimizing the cost of asset ownership. The results of the operational analysis must demonstrate whether or not the asset is meeting program objectives and the needs of the owners and users, as well as performing within baseline cost, schedule and performance goals.
- **Disposition Plan** (Final): The Disposition Plan is developed by the Project Team if the annual operational analysis concludes that the investment should be terminated. It involves either the transitioning of information, hardware, software, and documentation from the current system to another system, or the archiving or destroying of it. Each must ensure that the transition of the various components is done in an orderly fashion and ensures the confidentiality, integrity, and possible availability of the information in the future. It involves planning for the possibility of having to reinstall and bring the system back to an operational status, if necessary, and to preserve the data so it is effectively migrated to another system or archived for potential future access.
- **Architecture Models** – Business process, data, and technology/infrastructure models should exist that adequately describe the current state of this system and business usage.

Project Risks

- The original business needs supported by the system may have changed and/or the current solution may have not be truly or fully addressed those needs.
- Technology used in the original development of the solution may be close to become obsolete and/or unsupported which increase the risk for operational downtime and/or security vulnerabilities.

Major Probing Questions for the Enterprise Architecture (EA) Critical Partner

Use the following questions to assist you with completing your responsibilities for the stage gate review of this phase as appropriate to your subject matter expertise:

Question	**Evaluation Considerations**
1. *Has an Operational Analysis (OA) been conducted as required?*	Confirm that the rigor and quality of the OA process and documentation were appropriate for the size and scope of the system. Review and confirm that the OA indicates how well the functionality and capabilities of the system meet the organization's strategic goals, including the identification of gaps (if any), and plans for further developments, modernizations and enhancements (DME), to close the gaps and better meet the organization's strategic goals.
2. *Are the findings and recommendations of the OA consistent with previous assessments for this system?*	Request the most recent Investment Select assessment values for the system under review as well as any previous OA's. Check for consistency in these assessments.
3. *Does the system rely on any unsupported technologies? If so, what mitigation plans have been developed and are those plans*	Review the technologies being used. Identify technologies that are close to their "end-of-life" and may become unsupported. For each unsupported technology, confirm that mitigation plans are in place.

being followed?	

16. Disposition Phase: Role of an EA Critical Partner

The disposition activities ensure the orderly termination of the system and preserve vital information about the system so that some or all of the information may be reactivated in the future if necessary. Particular emphasis is given to proper preservation of the data processed by the system, so that the data is effectively migrated to another system or archived in accordance with applicable records management regulations and policies for potential future access.

Verify that any replaced/decommisioned system is correctly marked as decommissioned in the ESC. Also, verify that any dependencies or relationships to the expired system are redirected or similarly expired if no replacement capability exists.

Enterprise Architecture (EA) Critical Partner Anticipated Level of Effort

☒Low
☐Medium
☐High

Review Deliverables

Project Archives (Final): The Project Archives are the repository that preserves vital information, including both documentation of project execution and the data from the production system.

Project Risks

- Failure to fully plan for a systems retirement and disposition can result in problems within the architecture and especially with other systems that may be dependent on it.
- Failure to properly plan for the migration to a new system can result in data collection or data integrity issues.

Major Probing Questions for the Enterprise Architecture (EA) Critical Partner

Use the following questions to assist you with completing your responsibilities for this phase as appropriate to your subject matter expertise.

Question	**Evaluation Considerations**
1. *What are the impacts on dependent systems as a result of the decommissioning of this system?*	Validate that dependencies with other systems, stakeholders have been identified and addressed.
2. *Has the migration of capabilities and/or archival of data been well-planned?*	Confirm that plans for data migration and/or archiving are complete.

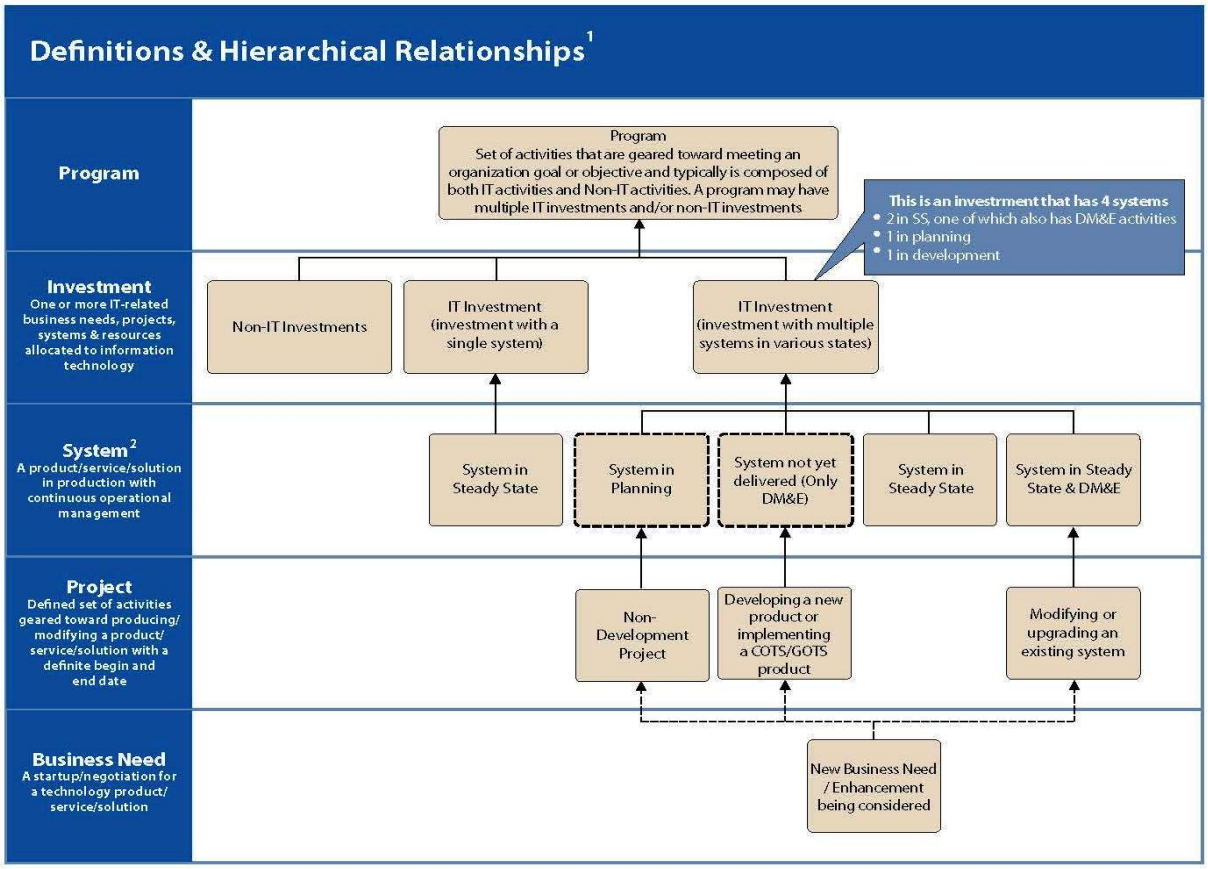
17.Appendix

Additional Resources

The following list is a composite of resources to assist in conducting Stage Gate Reviews:

- CDC Enterprise Architecture web site for CDC EA guiding principles (https://esp.cdc.gov/sites/ocoo/EITPO/EA/Pages/Default.aspx)
- CDC Information Technology Strategic Plan FY 2017 – 2021 for CDC IT goals https://esp.cdc.gov/sites/ocoo/EITPO/Documents/CDC%20IT%20Strategic%20Plan%202017-2021.pdf
- HHS Information Technology Strategic Plan Goals & Objectives (http://www.hhs.gov/about/strategic-plan/)
- CDC Enterprise Systems Catalog for inventory of existing CDC systems (http://esc.cdc.gov/)
- CDC Enterprise Critical Partners - visit the Critical Partners list on the EITPO site (http://esp.cdc.gov/sites/ocoo/EITPO/EPLC/Pages/cp.aspx).
- CDC EPLC – All EPLC documents - templates, job aids, etc. (http://esp.cdc.gov/sites/OCOO/EITPO/EPLC/Docs/Forms/AllItems.aspx)
- HHS EPLC Policy & Framework Documents (http://esp.cdc.gov/sites/ocoo/EITPO/EPLC/jobaid/Forms/AllItems.aspx)
- EITPO EPLC for definition and examples of documents and deliverables at all phases (http://esp.cdc.gov/sites/ocoo/EITPO/EPLC/Pages/framework.aspx)
- Federal C/I/O Council (https://C/I/O.gov/)
- Federal Enterprise Architecture web site for e-Government Initiatives and their architectures as described in the Federal Transition Framework (https://obamawhitehouse.archives.gov/omb/e-gov/fea /)
- CDC Enterprise Planning and Review Committee (http://esp.cdc.gov/enterprisesites/IRG/EPRC/default.aspx)
- CDC Information Resources Governance Council (http://esp.cdc.gov/enterprisesites/IRG/IRGC/default.aspx)
- CDC Financial Management Office (FMO) for budget formulation and appropriations guidance, budget execution, payments and executions, accounting, financial management systems, regulations/policies/procedures (http://intranet.cdc.gov/ofr/index.html) and service desk at ocfoservicedesk@cdc.gov
- CDC Office of Career Development (OWCD) for training considerations (http://intranet.cdc.gov/od/hcrmo/CDCUHome.shtml
- CDC policy and procedures related to Capital Planning and Investment Control (CPIC), including Earned Value Management (EVM) (http://esp.cdc.gov/sites/ocoo/EITPO/CPIC/default.aspx
- CDC Procurement and Grants (PGO) web site for contracts information and guidance (http://www.cdc.gov/od/pgo/funding/grants/grantmain.shtm) and Enterprise IT Program Management Office (EITPMO) (http://esp.cdc.gov/sites/ocoo/EITPO/Pages/Home.aspx)
- FIPS – 199 Minimum Security Requirements for Federal Information and Information Systems (http://csrc.nist.gov/publications/fips/fips200/FIPS-200-final-march.pdf)
- HHS Atlanta Human Resources Center (AHRC) for Civil Service HR considerations (http://intranet.cdc.gov/od/hcrmo/html/hr/index.html)
- HHS-OC/I/O Policy for Information Systems Security and Privacy Handbook (http://intranet.hhs.gov/it/cybersecurity/docs/policies_guides/PISSP/pol_for_info_sys_sec_and_priv_hndbk_9-22-2010.pdf)
- NIST – Special Publication 800-30 (Risk Management Guide for IT) (http://csrc.nist.gov/publications/nistpubs/800-30/sp800-30.pdf)
- OMB Circular A-11 for description of the Exhibit 300 and Exhibit 53 (https://www.whitehouse.gov/omb/circulars_a11_current_year_a11_toc)

Definitions & Hierarchical Relationships



[1] Operational support activities such as data preparation, analysis, extraction are not in scope for these definitions

[2] System refers to an IT System and not a set of business processes

[2] By definition, system is in a steady state unless the product/service/solution is new and has not yet been implemented

HSDP12

<u>Supporting HSPD-12 compliant Personal Identity Verification (PIV) Smart Cards</u>

As part of the federal HSPD-12 directive, all information systems deployed within any federal infrastructure must be able to support the HSPD-12 compliant PIV Smart Card that is issued to all federal employees and contractors. HSPD-12 (Homeland Security Presidential Directive-12) was issued in 2004 after the 9/11 attack when it was clear that federal agencies needed common identification standards and higher security. The HSPD-12 program is truly an enterprise architecture initiative to implement a common identity standard for all federal users.

Early inclusion of the HSPD-12 "identity architecture" in the planning and design phases will improve system interoperability and ensure that the HSPD-12 requirements are addressed. HSPD-12 compliance is mandatory requirement for continued federal funding starting in FY-12, per OMB memorandum M-11-11.

In accordance with these directives, the CDC project team's architect who is creating a new application (or recertifying an existing application) must ensure that PIV cards will integrate with the proposed software and hardware at the application level whenever the federal employee is being identified. In effect, user ID's and passwords are no longer considered a secure authentication method to access sensitive information by federal employees. The Security Critical Partner is charged with verification that the plan presented meets the requirement.

- For "Moderate" systems located inside the CDC firewall, a proxied authentication can be used. A CDC user can log in to the CDC network (which is a moderate system itself) with the PIV card and then the end-user system consumes the CDC network authentication.

- "High" systems must either use proxied authentication from a "high" system (like the HHS Access Management System, or AMS) or check the PIV card internally.
- "Low" Systems (often Internet facing) rated at EAL-1 with read-only access for federal users are not required to implement HSPD-12, also by OMB memorandum M-11-11.
- Some public facing and/or externally hosted government web applications may not require their general users to use a PIV card to authenticate, but may nevertheless require PIV logon for CDC users.

The following table may help to summarize the requirement and when it applies to systems owned and operated by the CDC. (The same rules apply to CDC-hosted or outsourced systems.)

System is physically located and accessed→ **and** **Target system is rated at ↓**	**Inside CDC network**	**Exposed to Internet**
Low **EAL-1, EAL-2**	"Kiosk" logon OK, no PIV needed for "read-only"	PIV not required for "read-only" access, User ID / Password OK. PIV required for making changes to content.
Moderate **EAL-2, EAL-3**	Network logon required. Proxied use of network logon OK. Direct PIV Logon OK.	Direct PIV Logon to system or proxied logon from moderate system required for read-access and for making changes to content.
High **EAL-3, EAL-4**	Direct PIV Logon to system or proxied logon from High system required	Direct PIV Logon to system or proxied logon from High system required for read-access and for making changes to content.

All critical partners must work with the security critical partner to ensure that HSPD-12 requirements are socialized and planned in the initial stages of the EPLC.

<u>Shared Authentication Services</u>

Shared Authentication Services are well developed at the CDC and should always be considered when there are communities of external users. Implementing CDC-approved methods of authentication (listed in the shared services catalog, such as SAMS and HHS-AMS) will be advantageous, as these services validate the government user's PIV card.

Shared Authentication Services remove the financial and programming burden of user ID and password management, and of PIV check, from the individual application team. In addition Shared Authentication Services provide the CDC external partner with a better CDC experience, as they can re-use their user credentials across many CDC programs that they are involved in. The combination may also result in cost savings in support and user maintenance, as help-desk resources are reused and not duplicated.

The Security Critical Partner is also providing verification that the plan presented has considered the shared authentication services requirement.